

Lesson 12: Independence and Conditioning

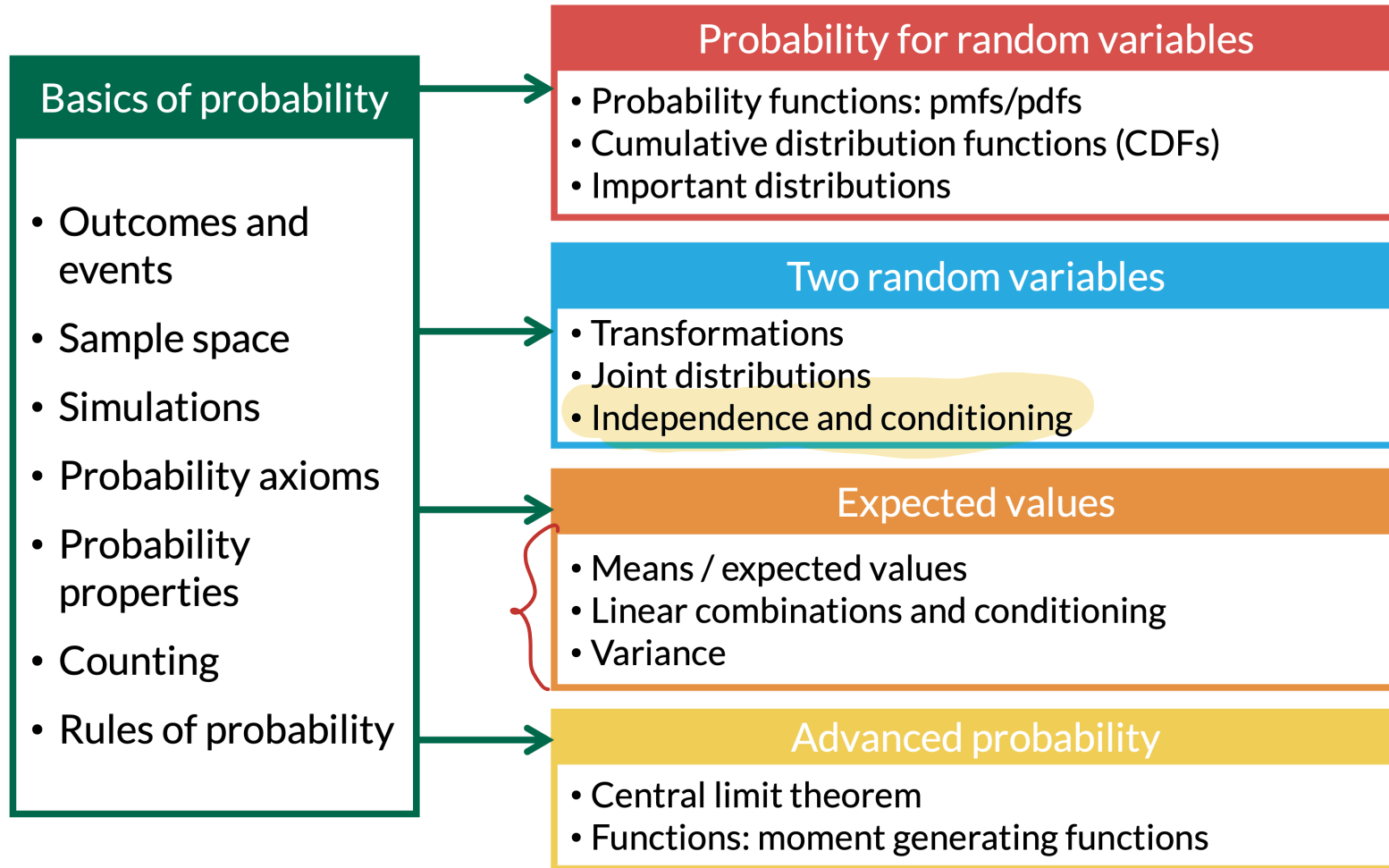
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Learning Objectives

1. Identify the formula for joint distributions for independent RVs and conditional distributions (PMFs/PDFs)
2. Find conditional pmf from a joint pmf and check if two RVs are independent.
3. Construct a joint distribution for two independent continuous RVs from their marginal distributions.
4. Calculate conditional probabilities and distributions for continuous RVs.

Where are we?



Learning Objectives

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How do we represent conditional pmfs/pdfs?

For events:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

For discrete RVs:

$$p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

$$p_{Y|X}(y|x) = P(Y = y|X = x) = \frac{p_{X,Y}(x, y)}{p_X(x)}$$

if denominator is greater than 0 ($p_Y(y) > 0$ or $p_X(x) > 0$)

For continuous RVs:

pdf $\neq$ prob

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

if denominator is greater than 0 ($f_Y(y) > 0$ or $f_X(x) > 0$)

How do we represent independent RVs in a joint pmf/pdf?

What do we know about independence for events?

For events: If $A \perp B$

$$P(A \cap B) = P(A)P(B)$$

$$P(A|B) = P(A)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

For discrete RVs: If $X \perp Y$

$$p_{X,Y}(x,y) = p_X(x)p_Y(y)$$

$$F_{X,Y}(x,y) = F_X(x)F_Y(y)$$

$$p_{X|Y}(x|y) = p_X(x)$$

$$p_{Y|X}(y|x) = p_Y(y)$$

For continuous RVs: If $X \perp Y$

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

$$F_{X,Y}(x,y) = F_X(x)F_Y(y)$$

$$f_{X|Y}(x|y) = f_X(x)$$

$$f_{Y|X}(y|x) = f_Y(y)$$

Remember: our probability rules must hold for these!

For discrete RVs

For a valid joint pmf, we need:

- $0 \leq p_{X,Y}(x, y) \leq 1$ for all x, y
- $\sum_{\{all\ x\}} \sum_{\{all\ y\}} p_{X,Y}(x, y) = 1$

For a valid conditional pmf, we need:

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For continuous RVs

For a valid joint pdf, we need:

- $f_{X,Y}(x, y) \geq 0$ for all x, y
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$

For a valid conditional pdf, we need:

- $f_{X|Y}(x|y) \geq 0$ for all x and y
- $\int_{-\infty}^{\infty} f_{X|Y}(x|y) dx = 1$

$$\int_{-\infty}^{\infty} f_{Y|X}(y|x) dy = 1$$

Extra notes

$$P(x_1, x_2, x_3) = P(X_1 = x_1) P(X_2 = x_2) \cdot P(X_3 = x_3)$$

- If $X_1, X_2, \dots, X_n$ are independent

$$p_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = \prod_{i=1}^n p_{X_i}(x_i)$$

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$$

$$F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) = \prod_{i=1}^n P(X_i \leq x_i) = \prod_{i=1}^n F_{X_i}(x_i)$$

- Don't forget, you can manipulate the conditional density to get the joint:

same goes for discrete RVs.

$$f_{X,Y}(x,y) = f_{X|Y}(x|y) f_Y(y)$$

$$f_Y(y) f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \times f_Y(y)$$

product of marginal CDF's

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4. Calculate conditional probabilities and distributions for continuous RVs.

Last class: joint distribution for two discrete random variables (1/2)

Example 1

Let X and Y be two random draws from a box containing balls labelled 1, 2, and 3 without replacement.

7. Find $p_{X|Y}(x|y)$.

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$$

$$\text{ex 1) } p_{X|Y}(X=1 | Y=1) = \frac{p(X=1, Y=1)}{p(Y=1)}$$

$$= \frac{0}{1/3} = 0$$

$$\text{ex 2) } p_{X|Y}(X=2 | Y=3) = \frac{p(X=2, Y=3)}{p(Y=3)} = \frac{1/6}{1/3} = \frac{1}{2}$$

$$p_{X|Y}(x|y) = \begin{cases} 1/2 & x \neq y \\ 0 & x = y \end{cases}$$

joint pmf:

		Y			
		1	2	3	$p_X(x)$
X	1	0	1/6	1/6	1/3
	2	1/6	0	1/6	1/3
	3	1/6	1/6	0	1/3
$p_Y(y)$		1/3	1/3	1/3	

Last class: joint distribution for two discrete random variables (2/2)

Example 1

Let X and Y be two random draws from a box containing balls labelled 1, 2, and 3 without replacement.

8. Are X and Y independent? Why or why not?

Remark:

- To show that X and Y are *not* independent, we just need to find one counter example
- However, to show that they *are* independent, we need to verify this for all possible pairs of x and y

	Y			
	1	2	3	$P_X(x)$
X	1	0	$1/6$	$1/3$
	2	$1/6$	0	$1/3$
	3	$1/6$	$1/6$	$1/3$
	$P_Y(y)$			
	$1/3$	$1/3$	$1/3$	

If $X \perp Y$:

$$P(X=x, Y=y) = \underline{P(X=x)} \underline{P(Y=y)}$$

$$\underline{x=1, y=1}$$

$$P(X=x, Y=y) = 0$$

$$P(X=1) = 1/3$$

$$P(Y=1) = 1/3$$

$$P(X=1, Y=1) = 0$$

$$\neq \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$$

$\Rightarrow X$ & Y are not independent

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Constructing a joint pdf from two independent, continuous RVs

Example 1.1

Let X and Y be independent r.v.'s with $f_X(x) = \frac{1}{2}$, for $0 \leq x \leq 2$ and $f_Y(y) = 3y^2$, for $0 \leq y \leq 1$.

1. Find $\underline{f_{X,Y}(x,y)}$

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

$$= \frac{1}{2} \cdot 3y^2 = \frac{3}{2}y^2$$

← when $X \perp Y$

$$f_{X,Y}(x,y) = \frac{3}{2}y^2 \quad \text{for} \quad \begin{matrix} 0 \leq x \leq 2, \\ 0 \leq y \leq 1 \end{matrix}$$

ex) domain w/ x dep on y

$$0 \leq x \leq y/2$$

immediately know $X \not\perp Y$

Probability from joint pdf from two independent, continuous RVs

Example 1.2

Let X and Y be independent rv's with $f_X(x) = \frac{1}{2}$, for $0 \leq x \leq 2$ and $f_Y(y) = 3y^2$, for $0 \leq y \leq 1$.

2. Find

$$\mathbb{P}(0 \leq X \leq 1, 0 \leq Y \leq \frac{1}{2})$$

$$\star f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

$$\mathbb{P}(0 \leq X \leq 1, 0 \leq Y \leq \frac{1}{2})$$

$$= \frac{1}{16}$$

$$f_{X,Y}(x,y) = \frac{3}{2}y^2$$

$$\text{OPT 1: } \mathbb{P}(0 \leq X \leq 1, 0 \leq Y \leq \frac{1}{2}) = \int_0^1 \int_0^{\frac{1}{2}} \frac{3}{2}y^2 dy dx$$

$$\text{OPT 2: } \mathbb{P}(0 \leq X \leq 1, 0 \leq Y \leq \frac{1}{2}) =$$

$$F(X=1, Y=\frac{1}{2}) - F(X=0, Y=0)$$

$$P(X \leq 1)$$

$$= F(X=1, Y=\frac{1}{2})$$

$$= F_X(X=1)F_Y(Y=\frac{1}{2})$$

$$= \left[\int_0^1 \frac{1}{2} dx \right] \left[\int_0^{\frac{1}{2}} 3y^2 dy \right]$$

$$= \left(\frac{1}{2}x \Big|_{x=0}^{x=1} \right) \left(y^3 \Big|_{y=0}^{y=\frac{1}{2}} \right) = \left(\frac{1}{2} - 0 \right) \left(\left(\frac{1}{2} \right)^3 - 0 \right) = \frac{1}{16}$$

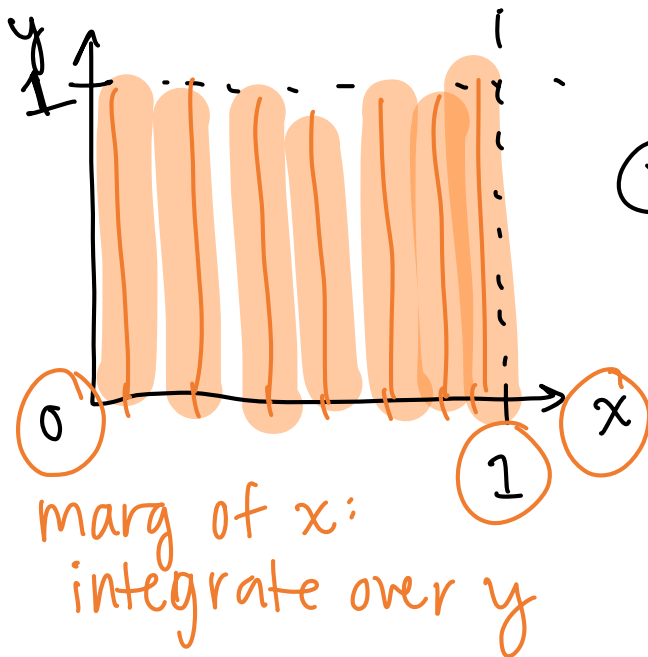
$$F = 0 \quad P(X \leq 0, Y \leq 0) = 0$$

Showing independence from joint pdf

Example 2.1

Let $f_{X,Y}(x,y) = 18x^2y^5$, for $0 \leq x \leq 1, 0 \leq y \leq 1$.

1. Are X and Y independent?



$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

$$\begin{aligned} \textcircled{X} \quad f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy = \int_{y=0}^{y=1} 18x^2y^5 dy \\ &= 3x^2y^6 \Big|_{y=0}^{y=1} = 3x^2(1)^6 - 3x(0)^6 \end{aligned}$$

$$f_X(x) = 3x^2 \text{ for } 0 \leq x \leq 1$$

$$\begin{aligned} \textcircled{Y} \quad f_Y(y) &= \int_{x=0}^{x=1} 18x^2y^5 dx = 6x^3y^5 \Big|_{x=0}^{x=1} \\ &= 6y^5 \text{ for } 0 \leq y \leq 1 \end{aligned}$$

$$f_X(x)f_Y(y) = [3x^2][6y^5] = 18x^2y^5$$

$$f_{X,Y}(x,y) = 18x^2y^5 = f_X(x)f_Y(y) \uparrow = \text{for } 0 \leq x \leq 1, 0 \leq y \leq 1 \Rightarrow X \perp Y$$

- ① Find marginals
- ② mult them together
- ③ look if equal to joint

Finding CDF from two independent RVs

Example 2.2

Let $f_{X,Y}(x,y) = 18x^2y^5$, for $0 \leq x \leq 1, 0 \leq y \leq 1$.

2. Find $F_{X,Y}(x,y)$.

$$\begin{aligned} \textcircled{3} \quad F_X(x) &= P(X \leq x) = \int_0^x f_X(s) ds = \int_0^x 3s^2 ds \\ &= s^3 \Big|_{s=0}^{s=x} = x^3 \text{ for } 0 \leq x \leq 1 \end{aligned}$$

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = \int_0^y f_Y(t) dt = \int_0^y 6t^5 dt \\ &= \dots = y^6 \text{ for } 0 \leq y \leq 1 \end{aligned}$$

STEPS

- ① given joint pdf
- ② get marg pdf
- ③ int marg pdfs to get CDFs
- ④ mult marg CDFs

$$\textcircled{4} \quad F_{X,Y}(x,y) = F_X(x)F_Y(y)$$

$$= \begin{bmatrix} 0 & x < 0 \\ x^3 & 0 \leq x \leq 1 \\ 1 & x > 1 \end{bmatrix} \begin{bmatrix} 0 & y < 0 \\ y^6 & 0 \leq y \leq 1 \\ 1 & y > 1 \end{bmatrix}$$

$$F_{X,Y}(x,y) = \begin{cases} 0 & x < 0 \text{ OR } y < 0 \\ x^3 y^6 & 0 \leq x \leq 1 \text{ \& } 0 \leq y \leq 1 \\ x^3 & 0 \leq x \leq 1 \text{ \& } y > 1 \\ y^6 & 0 \leq y \leq 1 \text{ \& } x > 1 \\ 1 & x > 1 \text{ \& } y > 1 \end{cases}$$

Showing independence from joint pdf

Do this problem at home for extra practice. The solution is available in Meike's video!

Example 3

Let $f_{X,Y}(x,y) = 2e^{-(x+y)}$, for $0 \leq x \leq y$. Are X and Y independent?

Final statement on independence

1. If $f_{X,Y}(x, y) = \underline{g(x)h(y)}$, where $g(x)$ and $h(y)$ are pdf's, then X and Y are independent.

- The domain of the joint pdf needs to be independent as well!!

transformations also independent

2. If $\underline{F_{X,Y}(x, y)} = \underline{G(x)H(y)}$, where $G(x)$ and $H(y)$ are cdf's, then X and Y are independent.

- The domain of the joint CDF needs to be independent as well!!

- Make sure that:

- X domain does NOT depend on Y
- Y domain does NOT depend on X

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Example starting from a joint pdf: first try!

Example 1.1

Let $f_{X,Y}(x,y) = 5e^{-x-3y}$, for
 $0 < y < \frac{x}{2}$.

1. Find

$$\mathbb{P}(2 < X < 10 | Y = 4)$$

Example starting from a joint pdf: second try! (1/2)

Example 1.1

Let $f_{X,Y}(x,y) = 5e^{-x-3y}$, for
 $0 < y < \frac{x}{2}$.

1. Find

$$\mathbb{P}(2 < X < 10 | Y = 4)$$

Example starting from a joint pdf: second try! (2/2)

Example starting from a joint pdf

Do this problem at home for extra practice. The solution is available in Meike's video!

Example 1.2

Let $f_{X,Y}(x,y) = 5e^{-x-3y}$, for
 $0 < y < \frac{x}{2}$.

2. Find $\mathbb{P}(X > 20 | Y = 5)$

Finding probability with conditional domain and pdf

Do this problem at home for extra practice. The solution is available in Meike's video!

Example 2

Randomly choose a point X from the interval $[0, 1]$, and given $X = x$, randomly choose a point Y from $[0, x]$. Find $\mathbb{P}(0 < Y < \frac{1}{4})$.